

T-cell and cutaneous lymphoma

Outcomes well behind B-cell disease — **PTCL is consolidated with transplant in first remission**, while **mycosis fungoides is managed for decades**. CD30 and blood involvement are the findings that change the plan
Stanford Oncology Handbook 2025, Lymphoma: Non-Hodgkin, pp.233–246 · NCCN Primary Cutaneous Lymphoma v2.2025 · items marked † are outside the handbook chapter

WHERE T/NK NEOPLASMS LIVE

Nodal	PTCL-NOS · AITL and nodal TFH lymphomas · ALCL, ALK+ or ALK- (DUSP22 / TP63 / NOS) · EBV+ nodal T-cell
Extranodal	Extranodal NK/T, nasal type · enteropathy-associated (EATL) · monomorphic epitheliotropic intestinal (MEITL) · hepatosplenic · subcutaneous panniculitis-like
Cutaneous	Mycosis fungoides · primary cutaneous CD30+ LPDs (lymphomatoid papulosis, C-ALCL) · primary cutaneous gamma/delta
Leukemic	Sezary syndrome · ATLL · T-PLL · T-LGL · aggressive NK-cell leukemia

WHO 5th edition 2024. **Sezary syndrome sits in both the cutaneous and leukemic columns** — that duality is the point: it is mycosis fungoides with blood involvement.

PIT — PERIPHERAL T-CELL LYMPHOMA

Factors	5-yr OS	10-yr OS
0	62.3%	54.9%
1	52.9%	38.8%
2	32.9%	18.0%
3–4	18.3%	12.6%

One point each: **age >60 · LDH > normal · ECOG 2–4 · marrow involvement**. Gallamini, Blood 2004. Staging otherwise uses Ann Arbor. **Compare these numbers with aggressive B-cell disease** — even no-risk-factor PTCL does worse than high-risk DLBCL.

MF/SS — ISCL/EORTC TNMB

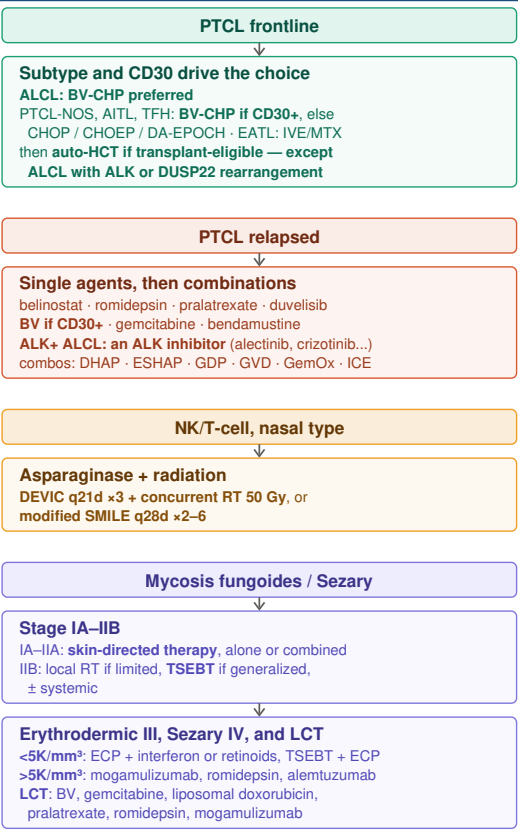
T1 · T2	Patches/plaques <10% BSA · >10% BSA (T2a patch only, T2b plaque)
T3 · T4	One or more tumors >1 cm · erythroderma ≥80% BSA
N1–N3	Clinically abnormal nodes by Dutch grade 1 / 2 / 3–4 (NCI LN 0–2 / 3 / 4)
B0	≤5% of lymphocytes are Sezary cells, or <250/μL, and <15% CD4+CD26– or CD4+CD7–
B1	>5% Sezary cells or >15% CD4+CD26–/CD7–, but not B2
B2	>1,000/μL Sezary cells or >1,000 CD4+CD26–/CD7– cells/μL with a blood clone matching the skin

Stage **IA** T1 · **IB** T2 · **IIA** T1–2 N1–2 · **IIB** T3 · **IIIA/IB** T4 with B0/B1 · **IVA1** = Sezary syndrome (any T, B2) · **IVA2** N3 · **IVB** M1. Olsen, Blood 2022.

PEARLS

- **Auto-HCT in first remission is the default for PTCL** — the exception is ALCL with an ALK or DUSP22 rearrangement, which does well without it
- **Allogeneic transplant is potentially curative in advanced MF/SS** — 1-yr and 3-yr OS 51% and 40%, PFS 42% and 33%, with acute and chronic GVHD 45% and 40%
- † **Sezary syndrome requires all three**: erythroderma (T4, ≥80% BSA), **B2** blood burden, and a **T-cell clone in blood matching the skin**. The older ISCL adjuncts — **CD4:CD8 ≥10** and loss of CD7 or CD26 — no longer stand alone; modified ISCL/USCLC/EORTC criteria need the clone **plus** an absolute count. The handbook prints none of this, giving only stage IVA1

TREATMENT PATHWAY



PTCL AND NK/T REGIMENS

Regimen	Dosing	Cycle
Pralatrexate	30 mg/m ² IV weekly ×6 — folic acid 1–1.25 mg daily from 7–10 d before until 30 d after, B12 1 mg IM q8–10 wk	q7 wk
Romidepsin · belinostat	Romidepsin 14 mg/m ² over 4 h d1, 8, 15 q28d ×6 · belinostat 1,000 mg/m ² over 30 min d1–5 q21d ×8	—
Duvelisib	75 mg PO BID ×2 cycles, then 25 mg PO BID	q28d, continuous
Gemcitabine · brentuximab	Gemcitabine 1,200 mg/m ² d1, 8, 15 q28d · brentuximab 1.8 mg/kg d1 (max 180 mg) q21d ×16 if CD30+	—
DEVIC	Dexamethasone 40 mg + ifosfamide 1,000 + mesna + etoposide 67, all d1–3; carboplatin 200 d1	q21d ×3 + RT 50 Gy
Modified SMILE	MTX 2,000 d1 · ifosfamide 1,500 + mesna + dexamethasone 40 mg + etoposide 100 d2–4 · pegaspargase 2,000 U/m² d7 , hold until MTX <0.04. Leucovorin from 24 h. TMP-SMX DS BID Sat/Sun	q28d ×2–6

CHOP/CHOEP/BV-CHP/DA-EPOCH dosing is **not in this chapter** — see the aggressive B-cell card.
EATL: CHOP ×1 then IVE (ifosfamide 3,000 + etoposide 200 d1–3, epirubicin 50 d1) alternating with MTX 1,500 d22, q28d ×3, then auto-HCT.

MF/SS THERAPIES

Therapy	Dosing and notes
Skin-directed	Topical corticosteroids · imiquimod · mechlorethamine (often compounded) · topical retinoids · local ISRT 8–12 Gy (24–30 Gy if unilesional) · TSEBT 12–36 Gy · PUVA for thicker plaques, NB-UVB for thinner
Bexarotene	300 mg/m ² PO daily — causes central hypothyroidism: follow free T4, not TSH ; check lipids. Vorinostat 400 mg PO daily · IFN alfa-2a 3–6 MU daily or TIW · ECP
Mogamulizumab	1 mg/kg IV d1, 8, 15, 22 then q28d — CCR4 antibody, best in Sezary/ blood disease . Mogamulizumab-associated rash is delayed, median onset 119 days
Romidepsin · brentuximab	Romidepsin 14 mg/m ² d1, 8, 15 q28d — QTc prolongation, replete Mg each infusion . Brentuximab 1.8 mg/kg q3wk — thicker skin disease and LCT only ; watch neuropathy
Methotrexate → pralatrexate	MTX 25 mg PO weekly (start 10–15, titrate); if higher doses are needed switch to pralatrexate 15 mg/m² d1, 8, 15 q28d — half the PTCL dose ; mucositis is dose-limiting
Alemtuzumab	3 mg d1 then 10 mg SQ on alternating days, TIW, max 12 wk — stage III/ T4 or Sezary only; CMV PCR q2–3 wk , Bactrim + acyclovir
Pembrolizumab	200 mg IV q3wk — continue indefinitely; stopping can lose the response

Maximise skin-directed therapy and supportive care first — it lengthens time to the next systemic treatment. Check compatibility with TSEBT/phototherapy first, and recycle agents that previously worked.

TRIALS

Trial	Role
ECHELON-2	BV-CHP vs CHOP in CD30+ PTCL — the reason ALCL gets brentuximab frontline
PROPEL	Pralatrexate in relapsed PTCL
BELIEF	Belinostat in relapsed/refractory PTCL
ALCANZA	Brentuximab vs physician's choice in CD30+ CTCL
MAVORIC	Mogamulizumab vs vorinostat in MF/SS — the registrational trial
Yamaguchi JCO 2009	DEVIC + concurrent RT in NK/T nasal type